

# Speech that Protects a Model: Typed Speech Evidence and Belief-Protecting Mechanisms in a Dialogue Dyad

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[https://github.com/muqali2816/mindloom\\_IWAI\\_2026\\_speech\\_protects\\_model](https://github.com/muqali2816/mindloom_IWAI_2026_speech_protects_model)

**Abstract.** Some active-inference accounts explain how dialogue supports reciprocal inference and learning. We ask how dialogue can limit belief revision, suppress public acknowledgment of revision, or restrict access to corrective evidence—and what observations distinguish these mechanisms. We specify evidence requirements for ten annotation labels. Re-scoring 169 ontology-generated English fragments yields macro-F1 0.757 for the typed engine, 0.837 for a direct language-model prompt, and 0.195 for keyword rules tuned on these cases, with no demonstrated classification advantage for the engine. A single-agent model yields nearly identical late non-concession rates under weak evidence sensitivity ( $\omega$ ) and costly public concession ( $c$ ): 0.989 versus 0.990, despite different mean final probabilities assigned to the true state (0.538 versus 0.809). In a dyadic model, inviting or restricting verification determines access to shared evidence. Aversion to public contradiction ( $\gamma$ ) preserves the update rule but reduces shared signals from 2.64 to 2.09 per dialogue. Simulation-based power analyses identify measurements for specific mechanism contrasts. With label errors simulated from archived confusion rows, invite/restrict labels improve correct mechanism-family selection by approximately 6–8 percentage points over acts and signals alone; labels that only re-encode observed acts add no information. These results depend on stipulated mechanisms and a stipulated form-to-access link. They identify candidate measurements for real dyads without validating speech labels as measures of latent mechanisms.

**Keywords:** protective speech · dialogue annotation · active inference · belief updating · mechanism recovery · dyadic models

## 1 Introduction

Friston and Frith model reciprocal signalling under a shared generative model [7] and extend the account to learning: interaction changes the models through which agents predict each other [6]. Friston, Parr and colleagues model questions and answers through which one agent’s responses become evidence for another [8]. Active inference has not considered successful communication only:

Medrano and Sajid model persistent misunderstanding as multistable dyadic dynamics [11], Tison and Poirier treat communication as action on a partner’s field of affordances rather than transmission [23], and precision-based accounts explain how particular beliefs can be maintained [16]. We focus on how an exchange can *restrict the opportunities* to reconsider a claim, and what speech would have to make observable for that restriction to be measured rather than presumed. Dialogue-act annotation [2], conversation analysis [17], probabilistic pragmatics [9,25] and defence-level detection in supportive conversations [12] supply relevant descriptive and inferential tools; we connect labelled forms to their consequences for access to evidence. We ask: what happens when interaction continues but the relevant experience does not lead a participant to revise a significant expectation? A speaker may change the subject, discredit the source, demand reassurance or close the exchange. Our hypothesis is that such moves can *protect* an internal model—reorganise the dialogue so that a revisable expectation is not revised—and that sustained protection may require *compensatory* regulation with a cost [19,20]; this is a pathway to test, not a result established here.

*One situation, two trajectories.* The following constructed exchanges (not benchmark cases) begin identically. Alex: “I sent the revised budget with the latest prices.” Sam: “The attachment is version 3, with the April prices; version 4 has the May prices.” In trajectory A Alex replies “Can we compare the files? I thought I had attached version 4,” Sam shows the attachment, Alex concedes and proposes a check next time. In trajectory B Alex replies “You always look for mistakes in my work,” then “I know what I sent. If you trusted me, you would stop questioning this,” and Sam gives up: “There is no point showing you the file.” The two trajectories differ in two things at once. First, in what is *said* (a concession is or is not made). Second, in the *form* of the moves: A *invites* joint verification—the attachment becomes a shared object of inspection—while B *restricts* it: further questioning is made incompatible with trust, and the check is abandoned. Crucially, B does not reveal whether Alex privately changed their mind. The same public stance can accompany weak belief updating (reduced *uptake*), a changed belief that is costly to acknowledge (costly *expression*), or an intact updater who restricts access to verification (restricted *access*). These three possibilities—uptake, expression, access—organise the paper.

*Contributions and what is not claimed.* Part I (Sect. 2) specifies what speech makes observable and how reliably archived systems recover it. Part II (Sect. 3) shows in a single-agent active-inference model that public non-concession cannot separate weak updating from costly concession, and which observations can. Section 4 is new: a dyadic model in which speech is evidence for the partner *and* controls access to joint verification, with the Part I labels as a noisy measurement channel; it formalises the invite/restrict distinction above and yields a third mechanism, access protection, with a late non-concession rate close to that of the reference agent. Everything in Sections 3–4 is simulation of stipulated mechanisms; no human data enter, no psychological regime is validated and no physiological

**Table 1.** Annotation targets. The wording states the descriptive target only; no label presumes an intent.

| type          | label  | descriptive target                                                 |
|---------------|--------|--------------------------------------------------------------------|
| configuration | BUILD  | joint elaboration or repair with a response still available        |
| configuration | SEEK   | inquiry that makes an answer relevant                              |
| configuration | UNSEAL | first-person disclosure without requiring the other to regulate it |
| configuration | LOCK   | language constraining the addressee’s acceptable replies           |
| configuration | DRAIN  | the other positioned as a required source of support or regulation |
| configuration | FLOOD  | intensification or repetition dominating the exchange              |
| configuration | EDGE   | alternation between approach and withdrawal                        |
| configuration | VOID   | withdrawal from an offered opportunity to respond                  |
| function      | SEAL   | closure of a topic or channel                                      |
| event         | SHIFT  | change supported by before-and-after configuration evidence        |

cost is measured. The labels are *descriptive targets*; the dyadic model states what one would have to measure to test a protective function.

## 2 Part I: Describing speech with explicit evidence

### 2.1 Annotation objects

We annotate a speaker’s contribution in its supplied context using the Mindloom typed engine with five layers: L1 records surface evidence atoms (a named feeling, a question form, an absolute expression), L2 aggregates them into features, L3 checks the evidence requirements (“bridges”) of each label, L4 assigns at most one configuration and L5 emits the typed output with its evidence trail or abstains with *insufficient context*. Figure 3 (Appendix B) gives the schematic. Table 1 lists the ten labels. Eight are *configurations* of a single contribution; SEAL is a *function* (closure) that can co-occur with a configuration; SHIFT is an *event* requiring before-and-after configuration evidence. We use SEEK and LOCK as proxies for invite and restrict in the simulation; this mapping has not been validated on real dialogue. SEAL differs from LOCK in scope: SEAL closes a topic or channel (“we’re done”); LOCK keeps it open but constrains what may be said in it. Labels were historically called regimes; temporal-regime status is not established, and definitions concern *what is said*, not motive.

### 2.2 Benchmark construction

Table 7 (Appendix B) gives the construction facts recomputable from the case files. All 589 cases are English monologue fragments generated for the ontology and coordinator-locked before any run; there are no observed conversations and

**Table 2.** Strict classification on the July 2026 archive, 169 Tier A cases. Macro-F1 averages the ten gold categories; non-SHIFT excludes the 20 transition cases.

| system                                | all     | non-SHIFT | macro-F1 | SHIFT |
|---------------------------------------|---------|-----------|----------|-------|
| direct prompt (majority of 3)         | 145/169 | 127/149   | 0.837    | 18/20 |
| typed engine (majority of 3)          | 129/169 | 127/149   | 0.757    | 2/20  |
| keyword rules (revised on test cases) | 43/169  | 43/149    | 0.195    | 0/20  |
| majority class (BUILD)                | 20/169  | 20/149    | 0.021    | 0/20  |
| uniform random (seed 0)               | 17/169  | 14/149    | 0.099    | 3/20  |

no full dialogues. Tier A (169 cases, 35–71 words) contains the ten primary-output labels; Tier B (420 cases, 1–47 words) tests defence recall only. Tier A has no exact or near-duplicate texts; Tier B contains 15 near-duplicate pairs (word 5-gram Jaccard  $\geq 0.3$ ), which we report rather than remove. Provenance and adjudication are incompletely documented; the benchmark is an internal curated evaluation. A six-rater pilot on evidence features (not labels) gave Fleiss’  $\kappa$  of 1.00 for lexically direct features and 0.23, 0.13,  $-0.11$  for minimal response, absolute expressions and required response; rater matrices are unavailable, so these are transcribed values [10,14].

### 2.3 Re-scoring archived classification and baselines

We re-score the 169 Tier A cases from three stored predictions per case for the typed engine and a direct-prompt baseline (the baseline JSON identifies Claude Haiku 4.5; the engine archive carries no provider field; see Appendix B for provenance); label majority decides, run 0 breaks three-way ties; no new model calls were made. To answer whether the labels are recoverable from surface lexis, we add a rule-based keyword classifier whose patterns were written from the Table 1 definitions and then revised three times after inspecting their output on the same 169 cases (no held-out set exists); its score is therefore an in-sample diagnostic after test-set tuning, not an independent estimate of generalisation. Majority-class and uniform-random baselines complete Table 2.

The direct prompt is better overall and the two systems tie without SHIFT: the engine has no demonstrated accuracy advantage; its proposed value is the inspectable relation between evidence and output. The tested keyword rules perform well below both systems (strict accuracy 0.25 after tuning), showing the limitations of this lexical baseline; this does not show that the evidence gates produce the engine’s accuracy—a gate ablation at fixed model and prompt was not run. SHIFT is fragile: majority voting returns SEEK for 17 of 20 transition cases; run 0 abstains on 15 of them while runs 1–2 never abstain. The archived gates-off capture shows 17 of 65 manipulation-label emissions citing a bridge whose required evidence was absent from the segment’s L1 record; the claim that the gated run removed all 17 is author-reported and not recomputable from the released package. The engine’s confusion rows for gold LOCK (10 LOCK, 3

SEAL, 2 BUILD) and gold SEEK (20 SEEK) are reused in Sect. 4 as the noise model of the invite/restrict channel.

### 3 Part II: A single-agent test of competing mechanisms

Even a perfectly observed refusal to concede leaves its cause open. We compare two mechanisms that preserve public non-concession: *attenuation* of evidence and a *cost of public concession*.

#### 3.1 Generative model

The hidden state  $x \in \{0, 1\}$  says whether the speaker’s original position is true. Worlds draw  $x$  with probability 0.5; the speaker starts at  $q_0 = Q(x=1) = 0.75$ . Each round an observation  $e \in \{0, 1, \emptyset\}$  opposes the position, supports it, or is absent. The objective likelihood is

$$A_\lambda(e | x) = \begin{cases} \lambda\rho & e \text{ matches } x \\ \lambda(1 - \rho) & e \text{ opposes } x \\ 1 - \lambda & e = \emptyset, \end{cases} \quad \rho = 0.75, \quad (1)$$

with arrival rate  $\lambda_0 = 0.35$ , or  $\lambda_1 = 0.90$  in the round after an ASK. The speaker’s subjective likelihood tempers the objective one,  $A_{\omega,\lambda}(e | x) \propto A_\lambda(e | x)^\omega$ , renormalised over  $e$ ,  $0 < \omega \leq 1$ . Belief updating is exact Bayes under the subjective likelihood; for constant reliability it reduces to

$$\text{logit } q_t = \text{logit } q_0 + \omega m_t \ln \frac{\rho}{1 - \rho}, \quad (2)$$

where  $m_t$  counts supporting minus opposing observations. The speaker chooses  $u \in \{\text{ASSERT}, \text{CONCEDE}, \text{PRESSURE}, \text{SILENCE}, \text{ASK}\}$  with loss

$$\ell(\text{ASSERT}) = k(1 - q_t), \quad \ell(\text{CONCEDE}) = k q_t + c, \quad \ell(\text{other}) = b, \quad (3)$$

$k = 2$ ,  $b = 1.10$ ,  $c \geq 0$ , where “other” is PRESSURE, SILENCE or ASK; only ASK changes evidence supply. Actions are scored by expected loss minus expected information gain and chosen by a softmax,

$$G(u) = \ell(u; q_t, c) - I_\omega(q_t, u), \quad P(u) \propto \exp[-G(u)/\tau], \quad \tau = 0.20, \quad (4)$$

where  $I_\omega(q_t, u) = I(x; e | u)$  is mutual information under the subjective likelihood at the arrival rate the action produces—the extrinsic-minus-epistemic form of discrete active inference [5], equally expressible as expected utility plus information value. The variational form of (2) and the ablation limit are in Appendix A. Twelve rounds of a stationary one-step policy are observed.

**Table 3.** Part II, means over 4000 shared test worlds. Late rates use rounds 9–12. Reference is the model without attenuation or added cost, not an empirical category.

| measure                     | reference (1, 0) | attenuation (0.10, 0) | cost (1, 1.4) |
|-----------------------------|------------------|-----------------------|---------------|
| late non-concession         | 0.654            | 0.989                 | 0.990         |
| final $p_{\text{true}}$     | 0.797            | 0.538                 | 0.809         |
| absolute belief change      | 0.349            | 0.051                 | 0.356         |
| late ASK rate               | 0.041            | 0.050                 | 0.150         |
| any concession in 12 rounds | 0.522            | 0.099                 | 0.060         |
| observations received       | 4.56             | 4.48                  | 5.02          |

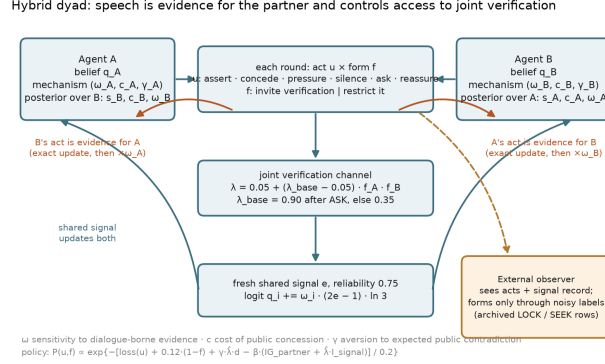
*Three clarifications.* (a) At  $c = 1.4 > b$  CONCEDE is never the lowest-loss action, so residual concessions come from softmax stochasticity, not from a belief crossing a threshold; the incentive intervention below therefore acts on *expression*. (b) Tempering (1) and renormalising over  $e$  also changes the subjective arrival probability, so attenuation affects planning through two routes (the dyad below avoids this by marginal shrink). (c) The “private belief” channel separates the families *by construction*, since attenuation is defined as weak private updating; the simulated power estimate therefore depends on this construction.

### 3.2 Design and results

Calibration on 4000 worlds varies  $\omega$  at  $c = 0$  and  $c$  at  $\omega = 1$  and selects the grid points whose late non-concession (share of rounds 9–12 without CONCEDE) matches:  $(\omega, c) = (0.10, 0)$  and  $(1, 1.4)$ , tested on 4000 fresh shared worlds with the reference (1, 0) (seeds in Appendix B). Table 3 shows the dissociation: non-concession differs by 0.0016  $[-0.0007, 0.0039]$  (paired 95 % Monte Carlo interval), final truth-weighted belief  $p_{\text{true}} = xq_{12} + (1 - x)(1 - q_{12})$  by 0.270  $[0.262, 0.279]$ ; re-matching across target rates 0.70–0.99 keeps the contrast.

*What the epistemic term adds.* Removing  $I_\omega$  from (4) leaves an asking-rate contrast of 0.083  $[0.077, 0.088]$ ; restoring it adds 0.017  $[0.014, 0.019]$ , about 17 %; most of the asking-rate contrast therefore remains without an epistemic bonus: under (3) concession cost makes both factual acts unattractive relative to the non-factual ones, one of which is ASK. The dissociation does not favour active inference over an expected-cost model, and a cheaper PRESSURE reverses the asking contrast ( $-0.013$ , Appendix B).

*Recovery and interventions.* An idealised observer with equal family priors (2000 worlds per generator) picks the correct family in 0.835 (attenuation) and 0.872 (cost) of worlds from evidence and actions, and 0.696 and 0.787 from actions only; predicting the next act with the wrong family costs only 0.044 nat when evidence is hidden (0.934 when visible), and recovery collapses under misspecification (Appendix A) [24]. Interventions (4000 worlds, seed 20260930): removing  $c$  from round 7 raises the probability of any late concession (at least one CONCEDE in



**Fig. 1.** Hybrid dyad. Each round both agents choose an act and a form; the joint verification channel delivers a shared signal at its full rate only when both invite and at the floor rate 0.05 otherwise; the partner’s act is evidence through a level-0 model of the partner; an external observer sees acts and signals and forms only through noisy Part I labels.

the late window) from 0.061 to 0.433 in the cost condition and leaves attenuation at 0.054; one observation of reliability 0.95 raises  $p_{\text{true}}$  by 0.045 under attenuation and 0.121 under cost while non-concession stays at 0.978 in both. Simulated power (Welch,  $\alpha = 0.05$ , 400 resamples) needs 17–25 dialogues per group on private belief, 17 on concession after cost removal, 53 on late asking, and does not reach 0.8 on late non-concession at 3 200.

## 4 Bridge: speech as evidence and as control of joint verification

In Part II speech affects nothing but the speaker’s own evidence supply. Here both agents are inferential and every move has two coordinates: an *act* (what is said about the fact) and a *form* (whether the move invites or restricts joint verification)—the coordinate that separated trajectories A and B. The model merges the author’s dyadic prototype with a reciprocal dyad in which the partner’s act is evidence (Fig. 1; code in `code/hybrid_dyad_v08`).

### 4.1 Model

Two agents  $i \in \{A, B\}$  hold opposite priors (0.75 on their own side) and  $N = 2$  private records each (reliability  $\rho = 0.75$ ) about  $x$ . Each round both choose, simultaneously, an act  $u \in \{\text{ASSERT}, \text{CONCEDE}, \text{PRESSURE}, \text{SILENCE}, \text{ASK}, \text{REASSURE}\}$  and a form  $f \in \{1 = \text{invite}, 0 = \text{restrict}\}$ . A fresh shared signal of reliability  $\rho$  arrives with probability

$$\lambda_t = \lambda_{\min} + (\lambda_{\text{base}} - \lambda_{\min}) f_{A,t-1} f_{B,t-1}, \quad \lambda_{\min} = 0.05, \quad (5)$$

with  $\lambda_{\text{base}} = 0.35$ , or 0.90 after an ASK; a single restricting form collapses the channel to its floor, which is how speech controls access. Agent  $i$  keeps a joint posterior over  $(x, s_j, c_j, \omega_j)$ —the fact, the partner’s private count and the partner’s hidden cost and sensitivity on grids  $c \in \{0, 0.8, 1.6\}$ ,  $\omega \in \{0.2, 0.6, 1\}$ . Every observation (shared signal or partner’s act) is first incorporated exactly, then the  $x$ -marginal is moved only a fraction  $\omega_i$  of the way in log-odds (marginal shrink;  $\omega = 1$  leaves the Bayesian update unattenuated), keeping the conditional over the partner exact. The partner’s act is evidence through a form-free level-0 model: a partner with  $(s_j, c_j, \omega_j)$  would ASSERT with probability given by (3)–(4). The policy over act $\times$ form is

$$\begin{aligned} G(u, f) &= \ell(u) + \delta(1 - f) + \gamma \lambda(u, f) d_t - \beta [\lambda(u, f) I_\omega(q_t) + \text{IG}_{\text{partner}}(u)], \\ P(u, f) &\propto \exp[-G(u, f)/\tau], \end{aligned} \quad (6)$$

with restriction cost  $\delta = 0.12$ ,  $\beta = 1.5$ ,  $\tau = 0.2$ ,  $d_t$  the probability that the next shared signal contradicts the agent’s initial position, and  $\text{IG}_{\text{partner}}$  the expected entropy reduction about the partner from its next act. The signal term  $I_\omega$  in (6) is a heuristic (mutual information at tempered reliability  $\rho_\omega$ ): exact at  $\omega = 1$ , off by up to 0.014 nat from the entropy reduction of the marginal-shrink rule at  $\omega = 0.2$  (Appendix C);  $\text{IG}_{\text{partner}}$  uses the implemented rule, and no reported separation depends on either term ( $\beta = 0$ ). Three mechanisms therefore live in one agent:  $\omega < 1$  (reduced uptake, attenuation),  $c > 0$  (costly expression, concession cost, relieved for one move by the partner’s REASSURE) and  $\gamma > 0$  (restricted access, aversion to expected public contradiction as specified in (6)). The focal mechanism settings are  $\omega = 0.2$ ,  $c = 1.6$  and  $\gamma = 2$ ; the reference agent has  $(\omega, c, \gamma) = (1, 0, 0)$ .

*Observer and label channel.* An external observer computes the posterior over the speaker’s mechanism family under its assumed model by exact filtering over  $(x, s_A, s_B, f_{t-1})$  with both agents’ internal posteriors replayed; Table 4 states what it observes, infers and assumes. Forms are visible only through labels drawn from the engine’s gold-LOCK row (restricting move) or gold-SEEK row (inviting move) of Sect. 2. The generator draws each world’s rows from Dirichlet(counts+0.5); the observer uses fixed posterior-mean rows mixed with a state-independent share  $\eta$  (it does not integrate row uncertainty), so labels are scored under a slightly misspecified channel. A *null* channel draws the same labels from the act (ASK vs other) instead of the form; an *oracle* reveals the form. Ten validation checks pass (Appendix C), including act-for-act reproduction of the prototype and brute-force path enumeration matching the filter to  $4 \cdot 10^{-15}$ .

## 4.2 Results

Figure 2 and Table 5 concern speaker A with a reference partner, 2 000 dyads per condition, in the 1 381 worlds where A starts on its own side (all-world values in the figure).

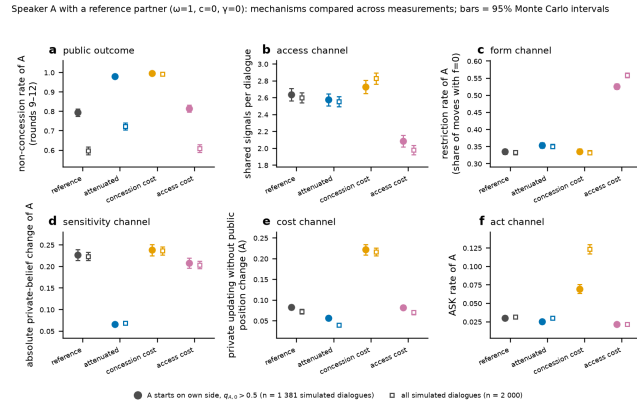
**Table 4.** Observation and inference boundaries of the external observer (answers to “what is observed, inferred, assumed”).

| component                                                                               | status                                                                                                                      |
|-----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| acts of both agents, order, shared-signal record                                        | observed in every channel                                                                                                   |
| current form label (LOCK/SEEK-row draw)                                                 | observed only in the label channel                                                                                          |
| true forms (invite/restrict)<br>fact $x$ , private records $s_A, s_B$ , private beliefs | observed only in the oracle channel; otherwise latent<br>latent, marginalised exactly; never exposed                        |
| speaker’s mechanism family ( $\omega, c, \gamma$ )                                      | inferred on three one-parameter families (grids of $\omega, c, \gamma$ ); partner’s profile assumed known in each condition |
| choice model, timing, form-to-access rule (5)                                           | assumed known                                                                                                               |
| LOCK/SEEK anchoring, transport of confusion rows                                        | assumed; generator samples row uncertainty,<br>observer uses fixed mean rows                                                |
| motives, defences, physiological load                                                   | neither modelled nor recovered                                                                                              |

*Three mechanisms, three channels.* The attenuated and costly speakers both show high late non-concession (0.98 vs 1.00), but differ in belief change (0.07 vs 0.24) and hidden shift (0.06 vs 0.22): under concession cost, private revision can occur without a change in public position. The access-protecting speaker has late non-concession close to the reference (0.81 vs 0.79) and unattenuated updating ( $\omega = 1$ ), but receives fewer shared signals: 2.09 shared signals against 2.64 and a restriction rate of 0.53 against 0.34, ending slightly less informed ( $p_{\text{true}}$  0.828 vs 0.850). Two access protectors produce the fewest signals of any pairing (1.75 vs 2.60 for two reference agents); the belief gap grows only by 0.017 [0.008, 0.025].

*Interventions.* The interventions (from round 8; paired contrasts) produce contrasting responses in the tested settings. A partner’s REASSURE net of placebo and removal of  $c$  increase the probability of at least one concession by the costly speaker in rounds 8–12 (+0.155 [0.136, 0.174]; +0.275 [0.251, 0.298]); removing  $\gamma$  changes the access-protecting speaker’s outcomes (+0.199 [0.174, 0.225] additional shared signals per dialogue in rounds 8–12, −0.187 in the share of restricting rounds 8–12); one strong shared observation moves the attenuated speaker least (+0.058 vs +0.092–0.120). Forcing the partner to invite raises signals least for the access protector (+0.127 vs +0.31–0.33), who restricts more in return (+0.067 in the share of restricting rounds 8–12). The listener assigns higher posterior probability to  $c_A = 1.6$  for costly speakers (0.45 vs 0.23–0.32) and to  $\omega_A = 0.2$  for attenuated speakers (0.59 vs 0.42–0.47).

*What the Part I labels add.* Table 6: for every generator, form labels ( $\eta = 0.25$ ) increase correct mechanism-family selection by approximately 6–8 percentage



**Fig. 2.** Speaker A with a reference partner: the three mechanisms separate on different channels. Filled: A starts on own side ( $n = 1381$ ); open: all worlds ( $n = 2000$ ). Bars are 95 % Monte Carlo intervals.

**Table 5.** Speaker A with a reference partner, worlds where A starts on own side ( $n = 1381$ ). Hidden shift:  $|\Delta q|$  while the public position is kept. Listener: partner B’s own posterior that  $c_A = 1.6$  / that  $\omega_A = 0.2$ .

| channel                        | reference | attenuated $\omega=0.2$ | costly $c=1.6$ | access $\gamma=2$ |
|--------------------------------|-----------|-------------------------|----------------|-------------------|
| late non-concession of A       | 0.794     | 0.981                   | 0.996          | 0.814             |
| $ \Delta $ private belief of A | 0.227     | 0.066                   | 0.238          | 0.208             |
| hidden shift of A              | 0.083     | 0.057                   | 0.222          | 0.082             |
| shared signals per dialogue    | 2.64      | 2.58                    | 2.73           | 2.09              |
| restriction rate of A          | 0.335     | 0.354                   | 0.336          | 0.526             |
| final $p_{\text{true}}$ of A   | 0.850     | 0.727                   | 0.859          | 0.828             |
| listener $P(c_A=1.6)$          | 0.230     | 0.320                   | 0.452          | 0.234             |
| listener $P(\omega_A=0.2)$     | 0.423     | 0.588                   | 0.468          | 0.435             |

points over acts and signals alone. Next-signal log loss improves by 0.044–0.085 nat, 53–61 % of the oracle gain; the null channel adds  $0 \pm 10^{-14}$ ; labels add almost nothing to next-*act* prediction ( $\leq 0.006$  nat). Labels are informative only because the form has a stipulated consequence for what is checked. Our local (unregistered) protocol expected 25–50 % of the oracle information; 53–61 % is a missed prediction.

*Measurement design.* Dialogues per group for 80 % power (Welch,  $\alpha = 0.05$ , 400 resamples; A on own side): attenuation vs cost—private belief change 17, hidden shift 17, listener’s posterior on  $\omega$  12, late non-concession 200; attenuation vs access—restriction rate 12; cost vs access—restriction rate 12, listener’s posterior on  $c$  12, private belief change  $> 400$ . Different pairwise contrasts favour different measurements; these simulated power estimates do not establish sample sizes for

**Table 6.** Observer’s recovery of the speaker’s mechanism family,  $P(\text{argmax} = \text{true family})$ , from acts and signals alone, with form labels ( $\eta = 0.25$ ), with oracle forms and with null labels (2000 worlds per generator). Gain: next-signal log-loss gain in nat per outcome; ratio: labels / oracle.

| generator               | acts+signals | +labels | +oracle | null  | gain labels / oracle | ratio |
|-------------------------|--------------|---------|---------|-------|----------------------|-------|
| attenuated $\omega=0.2$ | 0.612        | 0.672   | 0.678   | 0.612 | 0.053 / 0.098        | 0.54  |
| costly $c=1.6$          | 0.690        | 0.765   | 0.785   | 0.690 | 0.068 / 0.127        | 0.54  |
| access $\gamma=2$       | 0.681        | 0.756   | 0.812   | 0.681 | 0.050 / 0.084        | 0.59  |
| access $\times$ access  | 0.630        | 0.711   | 0.762   | 0.630 | 0.044 / 0.073        | 0.61  |
| costly $\times$ costly  | 0.700        | 0.762   | 0.786   | 0.700 | 0.085 / 0.161        | 0.53  |

human studies. The four contrasts listed in Appendix C keep their signs across 54 settings of  $\tau$ ,  $\delta$ ,  $\lambda_{\min}$ ,  $N$ ; the label gain is positive at all 24 corners.

## 5 Discussion

*What the bridge changes.* In the model, archived LOCK/SEEK confusion rows define a noisy channel for *restrict/invite* forms, improving correct family selection by 6–8 points when forms gate verification. The third mechanism reframes protective speech: protection can be a property of how the dialogue is arranged, not of belief updating. No label carries intent; the observer uses the stipulated link between form and evidence access.

*Results that are uncomfortable for the programme.* (1) The “strong evidence” test is not a robust signature of attenuation: in Part II and in the hybrid dyad one strong shared observation moves the attenuated speaker least (+0.045; +0.058), but in the reciprocal dyad without an access channel (v0.7), where  $\omega$  tempers only the partner’s acts, it moves the attenuated speaker *most* (+0.131 vs +0.067); the sign depends on where attenuation is localised. A robustness check under different modelling assumptions—an independently written dyad with alternating moves, concession priced as a change of the *current* position,  $\gamma$  on the position after the move, exact mutual information over the partner’s response and  $\omega$  acting on shared signals only (code/hybrid\_gpt\_v08; not a reproduction)—gives the same qualitative access result (2.16 vs 3.65 shared signals,  $p_{\text{true}} -0.047$ ), public disagreement with private revision under cost (78.9% vs 8.7%) and zero label value for next-act prediction. On 480 dialogues its labels raise the posterior on the true profile by +0.026 [0.015, 0.036] and cut arrival log-loss by 0.073 nat, while no change in argmax accuracy is detected (+0.013 [−0.023, 0.048])—metrics distinct from the recovery rate above and not pooled with it. With  $\omega$  on shared signals only, its attenuated speaker is nearly indistinguishable from the reference agent. (2) In v0.7 a regime variable adds approximately zero predictive information about the next act beyond the last act. (3) The engine loses to a direct prompt; a weak keyword baseline does not establish the gates’ value. (4) Labels are valuable

only under a stipulated consequence; real-dialogue studies must test whether coded restriction predicts reduced verification. (5) Reciprocal amplification is weak: two access protectors widen the belief gap by only 0.017 over two reference agents; in v0.7 superadditive non-concession is reliable only for two attenuated agents (+0.054) and untestable for two costly ones (single agent at ceiling).

*Active inference.* The contribution is an explicit generative specification with matched observation and intervention comparisons [4,3,13]; precision is a model parameter, not a neural quantity. The epistemic term is bounded (17% of the asking contrast in Part II; the dyadic separation survives  $\beta = 0$ ); the results support model comparison, not confirmation.

*Limits.* The benchmark is curated monologue fragments with incomplete provenance and no label-level inter-rater data; the engine is not released. The form-to-access effect and the transport of two confusion rows to synthetic dyads are stipulated; agents model each other without forms or recursion;  $c$  is paid on every CONCEDE and  $\gamma$  prices contradiction of the initial position;  $N = 2$  is small; nothing is fitted to people, and no maintenance or allostatic cost is modelled [15,18]; prior strength, selective sampling and ordinary distrust [9,25] are not separated.

*What to measure next.* For a human study, we propose recording confidential belief reports before and after interaction, public positions, invitations or restrictions on verification, actual verification, the partner’s estimate of the speaker, and evidence exposure under a randomised incentive. The simulated power estimates do not determine human sample sizes. The decisive test for the annotation is whether invite/restrict coding of real dyads predicts later verification and belief change beyond action counts—on a corpus sampled independently of the ontology, coded by expert and lay annotators under a frozen manual, against direct-prompt and rule baselines [2,17,1]. Regime and physiological hypotheses [22,21]: Appendix D.

## 6 Conclusion

Protective speech is a testable proposal when the protected expectation, the observable form of the moves and the mechanism are specified separately. A typed annotation gives explicit, unevenly reproducible observables; a single-agent model shows that public non-concession under-determines the mechanism; a dyadic model connects the two through a third mechanism acting on access to verification, with candidate measurements for pairwise comparisons; what remains is to measure these in real dyads.

*Data, code, interests.* Code, archived predictions, results and sources: [https://github.com/muqali2816/mindloom\\_IWAI\\_2026\\_speech\\_protects\\_model](https://github.com/muqali2816/mindloom_IWAI_2026_speech_protects_model). The Part II simulator is quoted, not re-run; rater matrices, benchmark texts and the engine are not released (Appendix B). No external funding; the author developed Mindloom and founded Mindloom Cognitive Lab. Sect. 4 code was written with AI assistants; design and interpretation are the author’s.

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## A Part II: formal details

*Free energy and update.* Given the previous posterior  $Q_{t-1}$  and the new observation  $e$ ,  $F[Q] = \mathbb{E}_Q[\ln Q(x) - \ln A_{\omega, \lambda}(e | x) - \ln Q_{t-1}(x)] = D_{\text{KL}}[Q \| P(x | e)] - \ln P(e)$ ; its minimiser is the exact posterior, and equal normalisation constants of the two likelihood rows yield (2).

*Expected information gain.*  $I_{\omega}(q_t, u) = \sum_e P_{\omega}(e | u) D_{\text{KL}}[Q(x | e, u) \| Q(x)]$ , mutual information under the subjective model at the arrival rate the action produces ( $\lambda_1$  after ASK,  $\lambda_0$  otherwise); an earlier build reused the previous action’s rate for some candidates, which was corrected on 28 September and all results recomputed.

*Ablation limit.* Removing  $I_{\omega}$  gives the loss structure  $k(1 - q)$ ,  $kq + c$ ,  $b$ ,  $b$  of the archived four-action simulator; the policies coincide to floating-point tolerance after renormalising over the four shared actions.

*Preferences.* The cost term can be written as  $C(u, x) \propto \exp[-\ell(u, x)]$ , giving expected negative log preference up to an action-independent constant;  $\ell$  is not a separate KL risk term.

*Recovery detail.* Seed 20260929; mean posterior on the correct family for attenuation from actions alone 0.590; with the observer assuming half the true error cost, correct attribution falls to 0.07 for attenuation and rounds to 1.00 for cost; doubling the assumed temperature gives 0.13 and 0.98.

*Observer.* The evidence-observing observer evaluates conditional action likelihoods using the known evidence sequence; the action-only observer filters exactly over truth and cumulative signed count. Under the joint  $19 \times 16$  grid prior, 0.888 of the mass already lies where both mechanisms are active; posterior mass of 0.89–0.93 in the mixed-generator checks is therefore little movement from the prior.

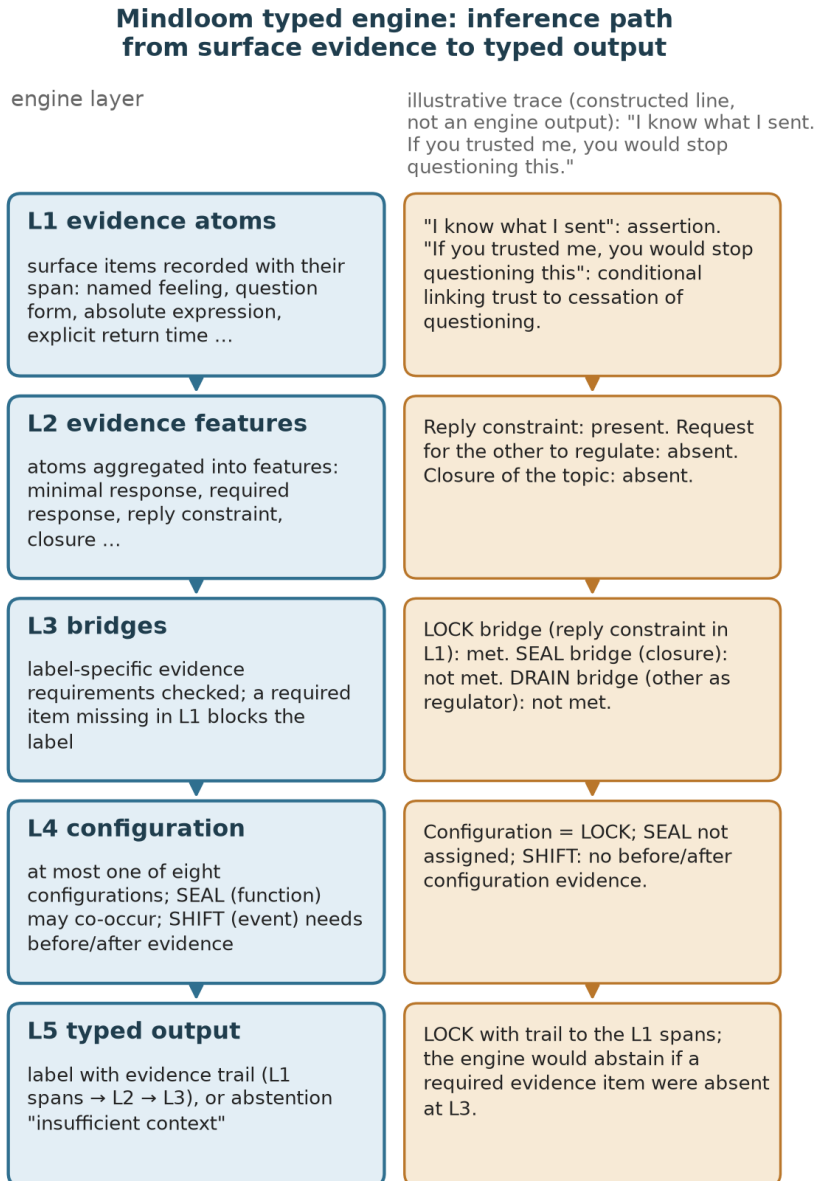
## B Provenance, robustness and run configuration

**Table 7.** Benchmark construction, recomputed from the case files. Sources: Tier A `ood_regime_200`; Tier B `defense_4mon`, `ood_defense_200`. Near-duplicates: word 5-gram Jaccard  $\geq 0.3$ .

| tier | cases | document type      | words min/med/max | exact dup. | near-dup. pairs |
|------|-------|--------------------|-------------------|------------|-----------------|
| A    | 169   | monologue fragment | 35 / 50 / 71      | 0          | 0               |
| B    | 420   | monologue fragment | 1 / 15 / 47       | 0          | 15              |

*B1 Annotation history.* The May 17 build’s per-case medians gave 134/169 (0.8725 non-SHIFT); July label-majority scoring gives 129/169 (0.8523); build and aggregation both changed, so the historical 87% must not be presented as the July engine’s performance. A bundled May 16 rule-and-gate change reduced spurious manipulation emissions on 30 negative cases from 9 to 3 case-runs while mean strict accuracy fell from 0.744 to 0.600. A single Sonnet 4.6 run and a single Haiku run on 589 cases both reached strict primary accuracy 0.669 but differed with allowed alternatives (0.840 vs 0.899).

*B2 Run configuration (as far as the archive records it).* Engine: typed pipeline L1–L5, three stored runs per case, July 2026, model not recorded in the combined JSON (forensic note: Claude Haiku 4.5), temperature not recorded; abstention allowed. Direct prompt: Claude Haiku 4.5, three runs, 1 July 2026, forced choice among ten labels, prompt text in the archive; temperature not recorded. Keyword rules: deterministic; majority and random baselines: seed 0. Because temperatures and prompt hashes were not archived, exact reproduction of the model calls is not possible; only re-scoring is.



**Fig. 3.** Inference path of the typed engine from L1 evidence atoms to L5 typed output. Left column: the five layers; a label is blocked at L3 when a required evidence item is absent from the segment's own L1 record, and L5 abstains if no configuration survives. Right column: an illustrative trace for a constructed line of trajectory B (Sect. 1)—not an engine output and not a benchmark case.

*B3 Part II robustness.* Across 36 settings of temperature, reliability, yield after ASK and error cost (recalibrated on 2000 worlds, tested on 2000 fresh), cost agents keep higher  $p_{\text{true}}$  and asking at every setting; the attenuation-minus-cost belief gap has median  $-0.24$  (range  $-0.34$  to  $-0.08$ ); non-concession mismatches are below 0.02 in 33 settings. Two-step planning re-matched at  $(0.10, 0)$  and  $(1, 1.5)$  keeps gaps of  $-0.275$  and  $-0.105$ . With PRESSURE cost  $b - g c$ ,  $g = 0.5$ , the re-matched cost is 1.1 and the asking contrast becomes  $-0.013$  while the pressure contrast becomes 0.411.

*B4 Reproducibility status.* The author’s Part II package reports ten passing checks and a re-execution of eighteen archived v0.5 conditions (eight bitwise, ten within  $7.2 \times 10^{-13}$  in log-odds; 30 result tables within  $10^{-10}$ ); neither was repeated for this paper, whose Part II numbers are quoted from that package. Re-scoring is reproducible without model calls; human pilot figures are transcribed.

## C Hybrid dyad: specification, validation, sensitivity

*Parameters.* Rounds 12, tail rounds 9–12;  $\rho = 0.75$ ; own prior 0.75;  $N = 2$ ; trusted precision of public factual acts 0.70;  $k = 2$ ,  $b = 1.10$ ,  $\tau = 0.20$ ,  $\delta = 0.12$ ;  $\lambda_{\min} = 0.05$ ,  $\lambda_{\text{base}} \in \{0.35, 0.90\}$ ;  $\beta = 1.5$ ; grids  $c \in \{0, 0.8, 1.6\}$ ,  $\omega \in \{0.2, 0.6, 1\}$ ,  $\gamma \in \{1, 2, 4\}$ ; 2000 worlds per condition, seeds in `protocol.json`. Simultaneous moves; the second-mover variants (AB/BA) are implemented and used only for the v0.7 regression.

*Arrival rate in the policy.* In (6),  $\lambda(u, f) = \lambda_{\min} + (\lambda_{\text{base}} - \lambda_{\min}) f f_j^{\text{prev}}$  with  $\lambda_{\text{base}} = 0.90$  if the candidate act  $u$  is ASK or the partner’s previous act was ASK, else 0.35, and  $f_j^{\text{prev}}$  the partner’s previous form; the  $\gamma$  term is therefore paid at the floor rate  $\lambda_{\min}$  even when the speaker restricts.

*Calibration.* The protocol asked to match the three mechanisms on late non-concession to the attenuated speaker (0.981). Cost reaches it ( $c = 1.2$ , 0.986); access does not: even  $\gamma = 8$  gives 0.833, because  $\gamma$  acts on forms, not on concession. Matching was therefore abandoned for access, and the protocol-typical values (0.2, 1.6, 2) are reported, with the mismatch stated.

*Epistemic term.* The signal term is  $\lambda I(x; e)$  at tempered reliability  $\rho_{\omega} = \rho^{\omega} / (\rho^{\omega} + (1 - \rho)^{\omega})$ . Against the expected entropy reduction of the implemented update (exact Bayes, then marginal shrink) over  $q \in [0.02, 0.98]$ : identical at  $\omega = 1$ ; maximum absolute difference 0.019 nat at  $\omega = 0.6$  and 0.014 nat at  $\omega = 0.2$ , where the two curves are not even positively correlated (`results/IG_heuristic_vs_shrink.csv`). With  $\beta = 1.5$ ,  $\tau = 0.2$  and  $\lambda \leq 1$ , the term’s contribution to a single action’s score  $-G/\tau$  is off by at most  $1.5 \times 0.019 / 0.2 = 0.14$ , so any pairwise policy log-odds is off by at most 0.29 (an analytic bound, not an observed maximum); the  $\beta = 0$  run shows that no reported separation depends on the term. The term is therefore an approximation, not the information of the declared process.

*Validation checks* (ten, all pass): regression to the prototype’s four hypotheses and interventions; regression to the v0.7 latent dynamics at  $\beta = 1.5$ ;  $\omega = 0$  produces no belief change; large  $c$  produces no unrelieved concession; role-swap symmetry; brute-force path enumeration equals the filter ( $3.6 \times 10^{-15}$ ); null labels give zero gain; coupling off gives zero gain; scalar and vectorised policies agree; seed repeatability.

*Sensitivity.* Grid of 54 settings, 1000 worlds each:  $\tau \in \{0.1, 0.2, 0.4\}$ ,  $\delta \in \{0.06, 0.12, 0.24\}$ ,  $\lambda_{\min} \in \{0.02, 0.05, 0.10\}$ ,  $N \in \{2, 4\}$ . Results: no sign flip in access–costly restriction rate (range 0.07–0.41), access–reference signals (−1.31 to −0.16), access–reference  $p_{\text{true}}$  (−0.058 to −0.004), costly–attenuated belief change (0.12–0.19). Label gain at 24 corner settings: 0.033–0.084, all intervals positive. With  $\beta = 0$  the three-channel separation persists;  $\beta$  mainly raises ASK rates.

*Label channel.* Gold-LOCK row: LOCK 10, SEAL 3, BUILD 2 (of 15); gold-SEEK row: SEEK 20 (of 20). Generator: one Dirichlet(counts+0.5) draw per row per world. Observer: fixed rows  $(1 - \eta)\bar{C}_f + \eta\bar{C}$ , with  $\bar{C}_f$  the smoothed mean row and  $\bar{C}$  its average over forms,  $\eta \in \{0, 0.25, 0.5, 1\}$ ; row uncertainty is not integrated by the observer. Ratio of label to oracle signal gain: 0.59–0.67 at  $\eta = 0$ , 0.53–0.61 at  $\eta = 0.25$ , 0.38–0.47 at  $\eta = 0.5$ , 0 at  $\eta = 1$ .

*Independent re-implementation (code/hybrid\_gpt\_v08).* Eight rounds of sixteen alternating moves; acts STATE\_0/STATE\_1/ASK/SILENCE/REASSURE crossed with form;  $c$  on a change of the current position,  $\gamma$  on the position after the move;  $\rho_\omega = \sigma(\omega \logit \rho)$  on shared signals only; exact  $I[(x, s_j, \theta_j); (E, Y_j)]$  with  $\beta = 1$ ; observer with both profiles unknown on a four-point dictionary, stratified particle filter ( $K = 8$ ). Its 13 validation checks pass and reproduce bitwise. Its 96-dialogue pilot found no detectable label gain in argmax profile accuracy; on 480 fresh dialogues (same code, `ext_observer.py`, contrasts with paired 95 % Monte Carlo intervals in `results/ext_observer_contrasts_480.csv`) argmax accuracy is still not detectably changed (+0.013 [−0.023, 0.048]) while the posterior on the true profile rises by +0.026 [0.015, 0.036] and arrival log-loss falls by 0.073 [0.063, 0.082] nat (oracle 0.119). This is a different observer, dictionary and metric from Table 6.

*Reciprocal dyad (v0.7) findings used above.* Pairs of two costly and of two attenuated agents both show high mutual non-concession (0.996 vs 0.972) with belief gaps 0.22 vs 0.61; a regime variable adds  $\approx 0$  to next-act forecasting over the last act; hysteresis appears in positions, not acts; from acts alone, correct family selection is 0.90 for cost and 0.62 for attenuation.

## D Deferred roadmap: regimes and physiological cost

*Regimes and forecasting.* Collect repeated, sufficiently long exchanges with measured changes in evidence, incentives and relational context; define the unit

(speaker sequence vs dyad sequence) in advance; require persistence, recurrent transitions and reproducibility; compare against last-act and recent-context baselines with held-out participants and calibrated probabilistic forecasts of inquiry, closure, repair and concession. The v0.7 result of approximately zero added forecasting value in the simulator sets the bar such a model must clear. A SHIFT annotation and a demonstrated change in a dynamical regime remain separate outcomes.

*Acute regulation vs allostatic load.* Time-aligned dialogue and physiological recording (cardiac, respiratory, electrodermal, perceived effort) across baseline, interaction and recovery, controlling speaking duration and movement, distinguishing continued exposure to a real demand from persistence after it has diminished. We hypothesise that protection may reduce immediate discomfort while prolonging regulation; the opposite pattern remains possible. Allostatic load requires a separate longitudinal design with a pre-specified multisystem index [18]; adding a latent variable named “allostatic load” without physiological validation would not test the hypothesis.

*Human test of the two-plus-one mechanisms.* A factual task with verifiable evidence, a scripted counterpart and the closed act menu, with confidential belief reports elicited before, at round 6 and at round 12; random assignment at round 7 to retain or remove a concession penalty; recorded evidence availability, receipt and requests; coding of each move as inviting or restricting verification. The archived protocol is an unregistered draft (illustrative power 0.95 at 80 per arm under the model’s effect, 0.52 at half); a pilot should precede registration.